Shared Aperture Waveguide Module

Theory and Design Reference



2026-04-07

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1. Introduction

The design of high output, wide-bandwidth loudspeaker systems with controlled directivity presents a set of competing constraints in acoustic, geometric, and ergonomic domains. Conventional multiway direct radiating systems are often limited by spatial separation between sources, which leads to frequency dependent interference and off-axis lobing through the crossover regions. While careful crossover design and driver placement can mitigate these effects, they cannot eliminate the underlying issue of distributed acoustic origins.

Coaxial topologies address this limitation by approximating a coincident source, improving phase coherence and polar consistency. However, such systems introduce their own constraints, including limitations in scalability, amplitude modulation distortion, and achievable bandwidth partitioning across transducers. In high-output applications, these constraints can become prohibitive.

Multiple entry horn (MEH) and related waveguide architectures offer an alternative approach by acoustically summing multiple sources into a shared radiating aperture. When properly implemented, these systems can approximate a point source while maintaining high efficiency and controlled directivity. However, practical implementations are constrained by geometric requirements - particularly the need to maintain sufficiently small source-to-source spacing relative to the wavelengths at crossover frequencies, while simultaneously preserving favorable waveguide expansion near the throat.

This document introduces the Shared Aperture Waveguide Module (SAWMOD), a compact three-way loudspeaker topology that integrates high frequency and midrange transducers into a unified waveguide structure with an augmented low frequency section featuring a passive side/rear rejection mechanism. The design incorporates a unique waveguide geometry featuring localized contour elements (“knuckles”) that enable reduced effective source spacing for bandpass-coupled midrange drivers while maintaining controlled wavefront expansion.

The presented mechanisms of this paper are:

- A waveguide topology that relaxes conventional geometric constraints associated with multiple entry horn designs
- A method for improving acoustic source spacing without excessive disruption to throat expansion
- A parametric contouring approach that enables targeted directivity control through wavefront shaping
- An integrated low-frequency system that extends pattern control to lower frequencies through spatial and acoustic interaction mechanisms

The following sections describe the design objectives, geometric formulation, and acoustic behavior of the SAWMOD system, supported by simulation and measurement based analysis and experimental observations.

2. Acoustic Objectives

The system is designed as a three way architecture with defined crossover regions chosen to balance bandwidth allocation, directivity control, and transducer performance:

- Low-Mid Crossover: ~350 Hz
- Mid-High Crossover: ~1.6 kHz

A key objective is to maintain consistent directivity through these crossover regions. In particular, the mid-high transition imposes a stringent constraint on allowable spatial separation between acoustic sources. As a design guideline, the effective center-to-center spacing between radiating elements should remain on the order of ≤¼ wavelength at the crossover frequency to minimize off-axis interference. For a 1.6 kHz crossover, this corresponds to approximately 53.5 mm.

The target nominal coverage pattern of the system is:

- Horizontal: 100°
- Vertical: 70°

Maintaining this coverage with minimal frequency-dependent narrowing or lobing is a primary design goal. Achieving the desired acoustic performance within a compact form factor introduces several geometric challenges:

2.1 Waveguide Throat Region

- The waveguide must support smooth wavefront expansion from the high frequency compression driver while accommodating the injection of midrange energy. Conventional approaches often require placing midrange injection ports close to the throat to satisfy spacing constraints, which can disrupt the initial expansion and introduce acoustic discontinuities.

2.2 Midrange Source Placement

Bandpass-loaded midrange drivers must be positioned such that:

- Their effective acoustic centers remain within the spacing constraint at the crossover frequency
- Their coupling into the waveguide minimizes high-Q resonances and excessive diffraction

2.3 Aperture Development

- The waveguide must transition from a compact throat to a rectangular mouth geometry suitable for pattern control and arrayability (for future designs sharing this topology), while preserving a controlled directivity profile across the operating bandwidth.

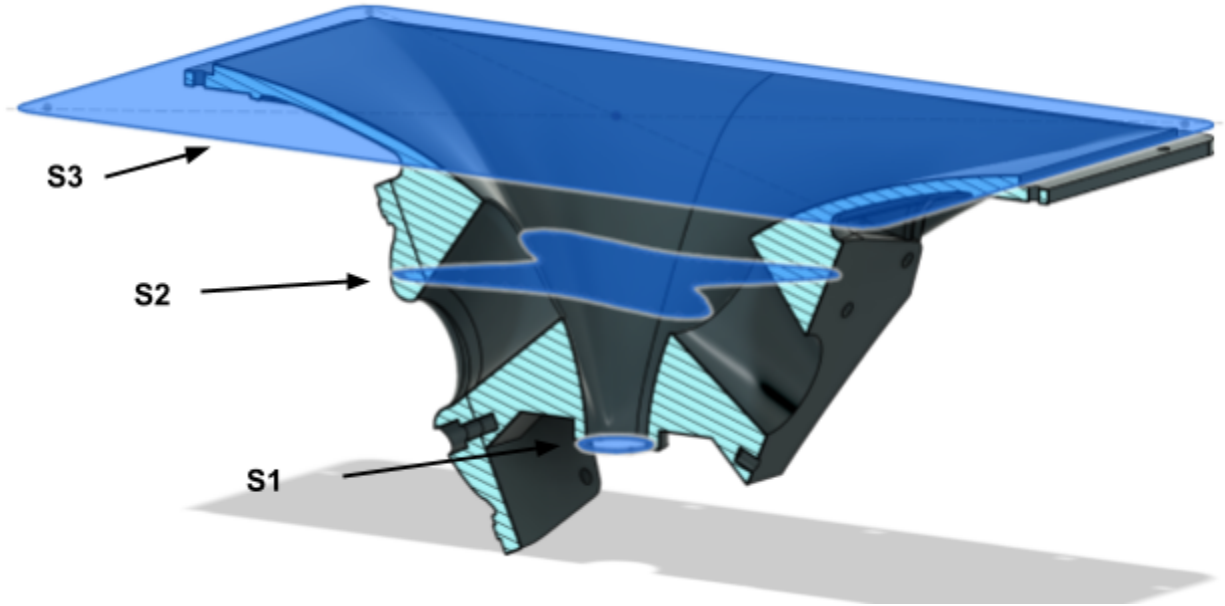
The integration of low-frequency transducers introduces additional requirements:

- Extension of directivity control below the midrange band
- Efficient use of enclosure volume within a compact footprint
- Thermal and mechanical considerations for high power operation

The selected configuration utilizes dual sidefiring low frequency drivers coupled to a shared enclosure volume, with spatial arrangement chosen to influence the system's polar response near the low-mid crossover region.

3. Waveguide Topology

For the purpose of this document, this waveguide topology is referred to as a three segment device consisting of axial cross section segments denoted as S1 (the throat), S2, (the segment located at the centroid of the bandpass injection ports), and S3 (the segment at the mouth).

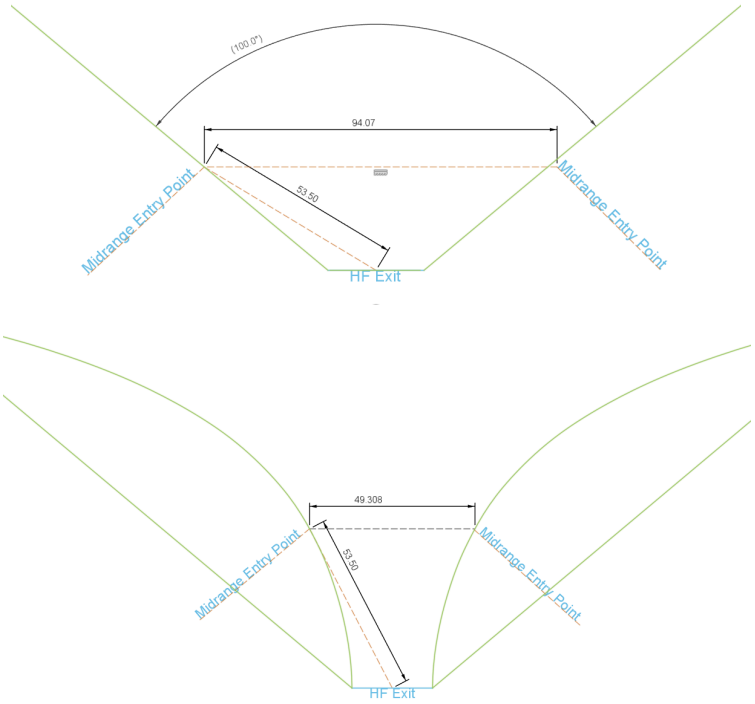


The SAWMOD waveguide incorporates localized inward-extending contour features at the S2 region, referred to as knuckles. These features modify the cross sectional geometry of the waveguide in a manner that is not captured by traditional monotonic flare descriptions. Rather than uniformly expanding from throat to mouth, the waveguide introduces controlled local deviations that alter both the spatial distribution of the boundary and the effective propagation path of the wavefront. The knuckles serve two primary geometric functions:

1. **Reduction of Effective Source Spacing**

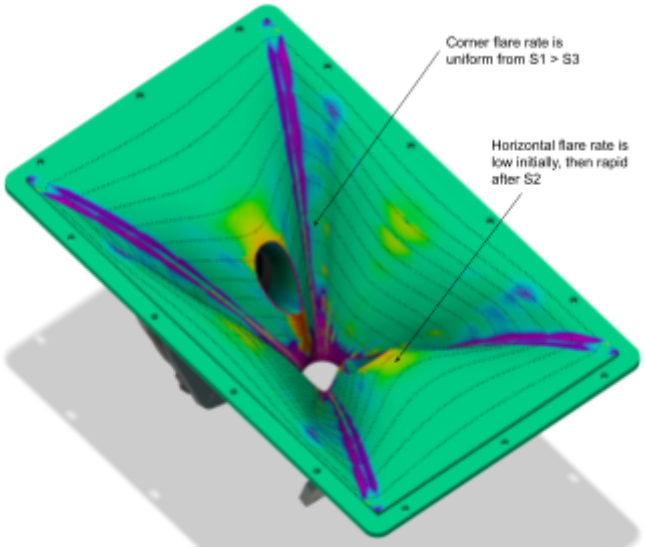
By extending inward from the waveguide walls, the knuckles allow the midrange injection ports to be positioned closer to the central axis without requiring a corresponding reduction in the overall cross-sectional area at S2. This enables reduced center-to-center spacing between the midrange apertures and the high-frequency source, improving conditions for constructive summation through the crossover region.

The illustration on the right displays the center-center bandpass injection port distances comparatively between a conical multiple entry horn and the SAWMOD having the same horizontal coverage.



2. **Modification of Local Expansion Rate**

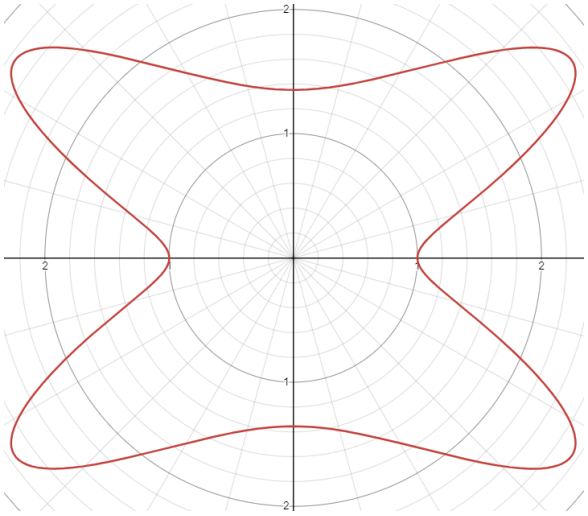
The presence of the knuckles alters the local rate of cross-sectional expansion in the vicinity of S2. Instead of a purely monotonic flare, the waveguide exhibits a compound expansion profile in which the apparent flare rate is locally reduced along the primary axis while remaining consistent with the global aperture growth. Moreover, the parametric model used to define the shape of the device at S2 provides an additional degree of freedom for shaping the polar response, particularly in frequency regions where the bandpass injection ports create pattern narrowing.



3.1 Waveguide Contour Methodology

The waveguide topology consists of a compound contour that supports uniform directivity development along the primary horizontal axis, and allows for compensatory pattern widening in the lower treble to alleviate the polar narrowing effects of the midrange bandpass injection ports[1]. The profile was developed in Ath4 - a parametric script-based waveguide mesh generation tool. The cross-sectional bounding shape of the waveguide at S2 is driven by a superformula with the following parameters:

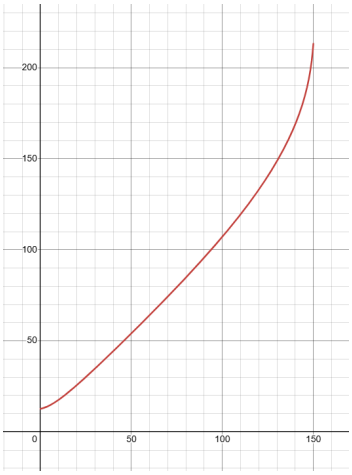
$$r(\theta) = \left[\left| \frac{\cos\left(\frac{4\theta}{4}\right)}{1} \right|^{14} + \left| \frac{\sin\left(\frac{4\theta}{4}\right)}{1.2} \right|^{2.5} \right]^{-\frac{1}{1.5}}$$



The superformula approach introduces a large degree of freedom in defining the aperture shape and resulting directivity characteristics. To define a curve that creates a smooth transition of the geometry axially, an implementation of the OS-SE profile is used:

$$y = \sqrt{(kr_0)^2 + 2kr_0 x \tan\alpha_0 + (x\tan\alpha)^2} + r_0(1-k) + \frac{Ls}{q}\left(1 - \left(1 - \left(\frac{qx}{L}\right)^n\right)^{\frac{1}{n}}\right) \quad \{0 \le x \le L\}$$

L = 115
s = 0.8
q = 0.996
n = 4
r = 12.7
α_0 = 0
k = 1



[1] https://www.diyaudio.com/community/threads/compact-almost-fullrange-meh.430398/post-8160276

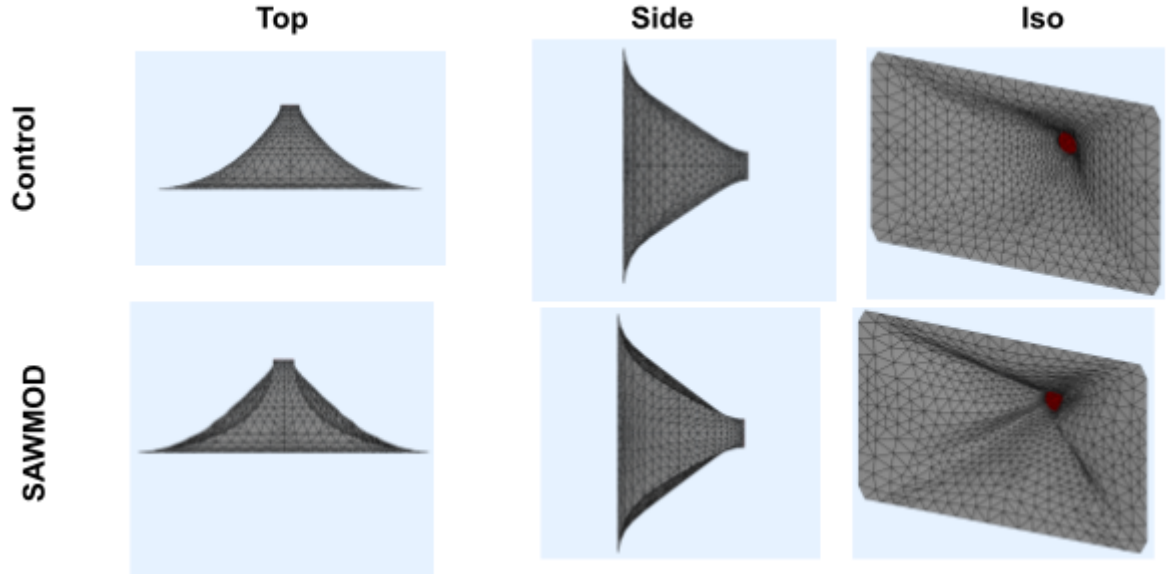
Ath4's rectangular morph interpolation feature is used between S2 and S3 to gradually transition to a rectangular exit with a morph rate of 4.

While the horizontal cross section of the SAWMOD features a curve resembling an exponential profile, the waveguide as a whole cannot be accurately described as exponential as it features a dual-contour hybrid profile defined by the knuckles and valleys. By adjusting parameters in the superformula, OS-SE curve, and morph rate, the depth and width of the knuckle features can be controlled, having an effect on the wavefront shape as it progresses through the device. Functionally, the knuckles modify the spatial rate at which boundary conditions influence wavefront curvature.

- This aids in the creation of a rectangular-bounded wavefront shape at the mouth of the device.
- Allows for more control in defining horizontal beamwidth in the 3-7kHz region to compensate for polar narrowing effects of bandpass injection ports.

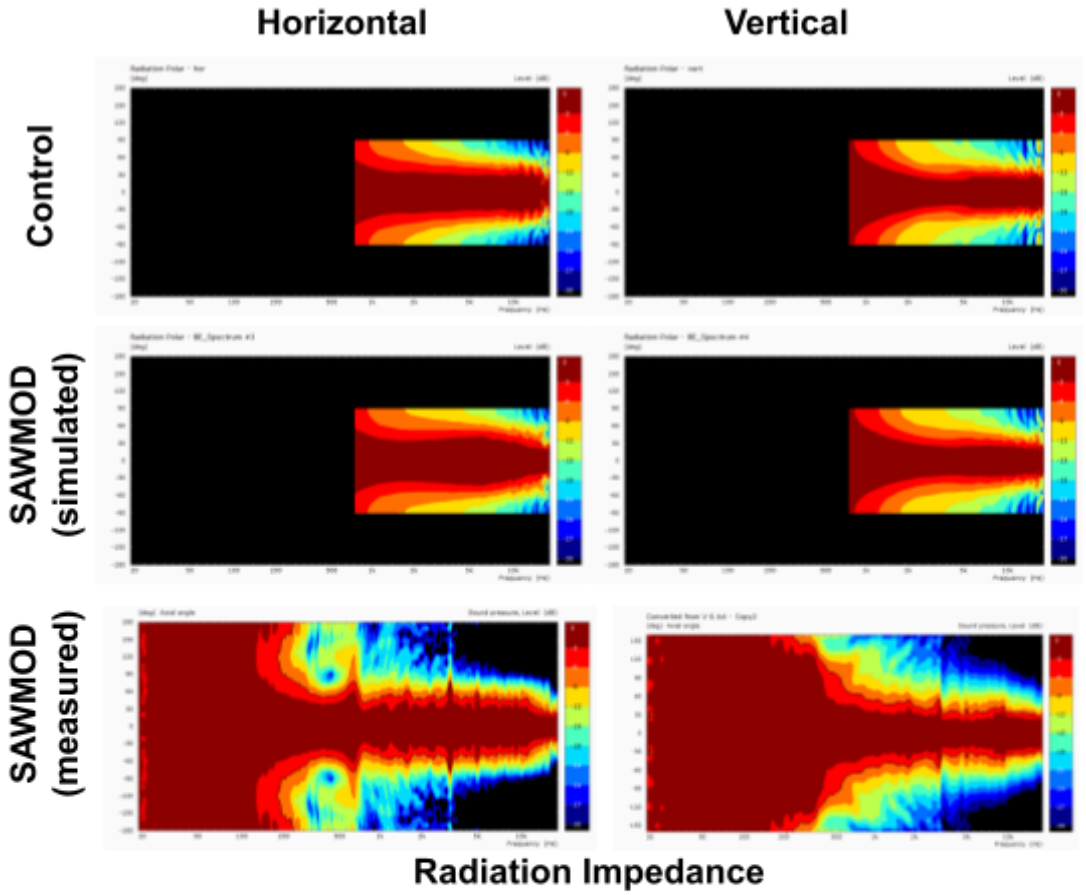
4. Wavefront Simulation Analysis

The following field analyses were performed in AKABAK. A control waveguide was generated using Ath4 having the same OS-SE guiding curve parameters as the SAWMOD. Axial length, throat entrance angle, and mouth dimensions are identical between the two candidates for a comparative analysis.

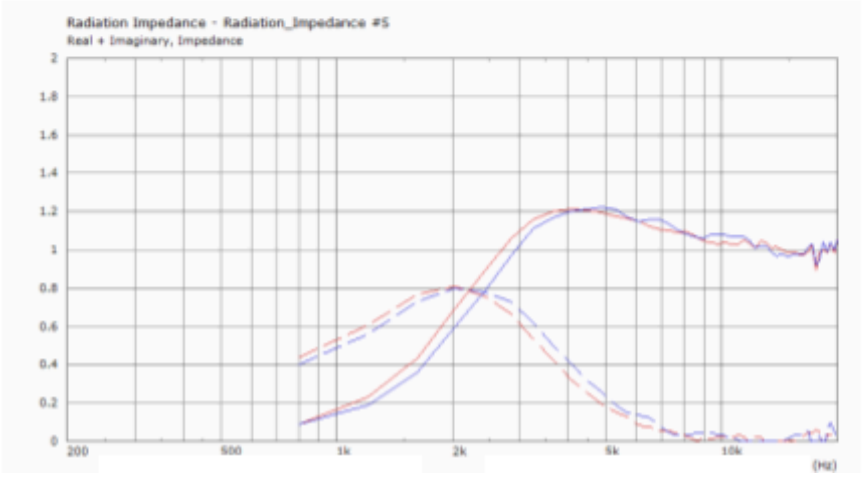


The compression driver was modeled as an idealized uniform velocity source at the waveguide throat, isolating the effects of waveguide geometry from phase plug-specific behavior. The simulation was configured as a half-space infinite baffle arrangement. Axial planar slices were positioned at S2, S2.5, and S3 to illustrate the wavefront shaping effects of the knuckles

4.1 Directivity & Radiation Impedance Comparison

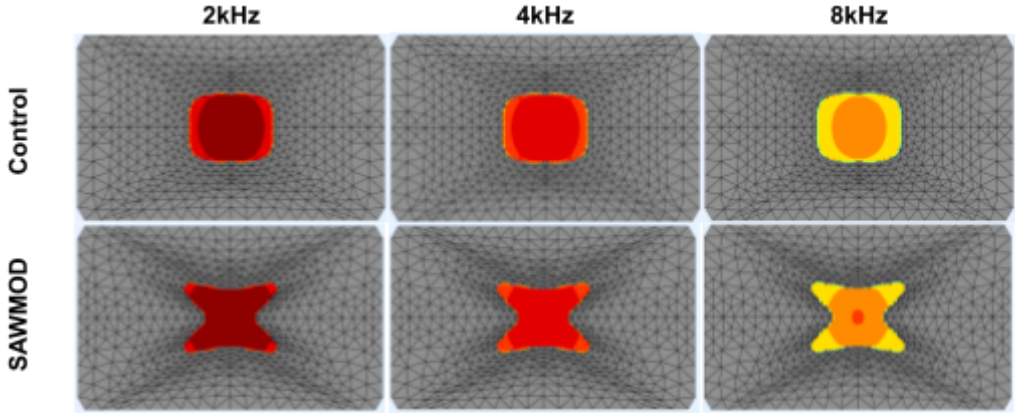


Radiation Impedance

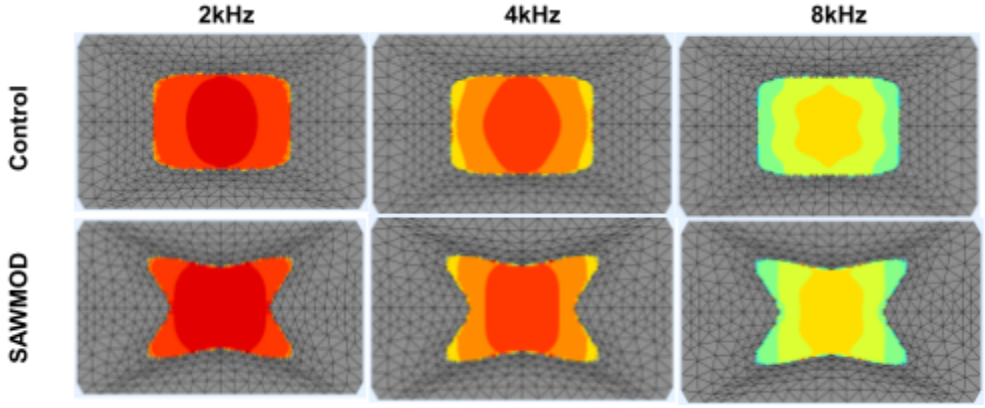


4.2 Sound pressure field analysis (3dB contours)

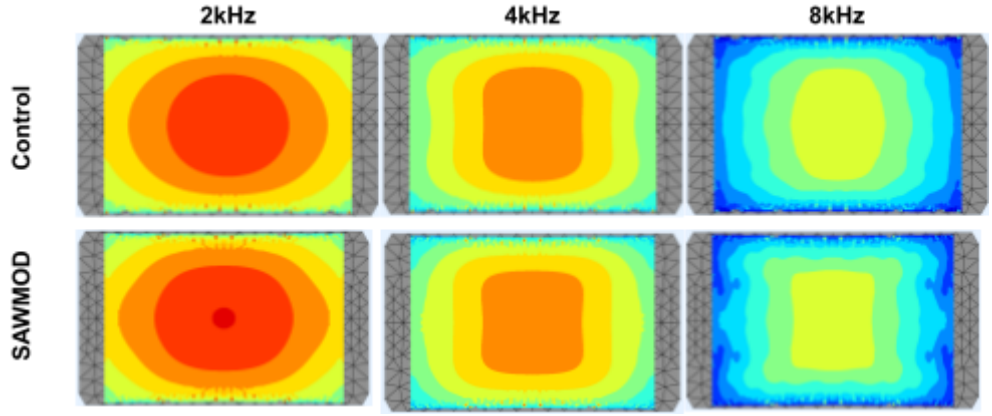
S1 (53.5mm axial offset)



S2.5 (84.25mm axial offset)



S3 (115mm axial offset)



4.3 Simulation Results

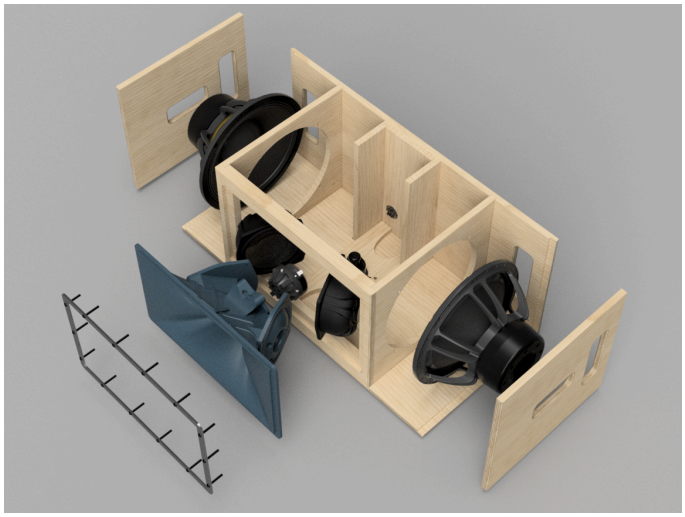
Performance between the two variants is largely similar, with a few notable differences:

1. The additional degree of freedom afforded by the knuckle features allows for a compensatory pattern widening from 3-7kHz. The comparison between simulated and measured indicates that the pattern was successfully optimized for constant directivity.
2. Radiation impedance has increased slightly.
3. The wavefront shape at the mouth is largely similar but with a slightly more rectangular-bounded pattern in the HF.

The SAWMOD waveguide departs from conventional monotonic flare geometries by introducing controlled, localized contour features that serve both geometric and acoustic functions. By doing so, it addresses a central limitation of multiple entry horn designs: the tradeoff between source spacing and waveguide expansion integrity.

The resulting topology enables closer effective spacing of midrange injection sources, controlled wavefront evolution through critical regions of the device and additional flexibility in shaping directivity without increasing overall system size

5. Low Frequency Transducer Integration

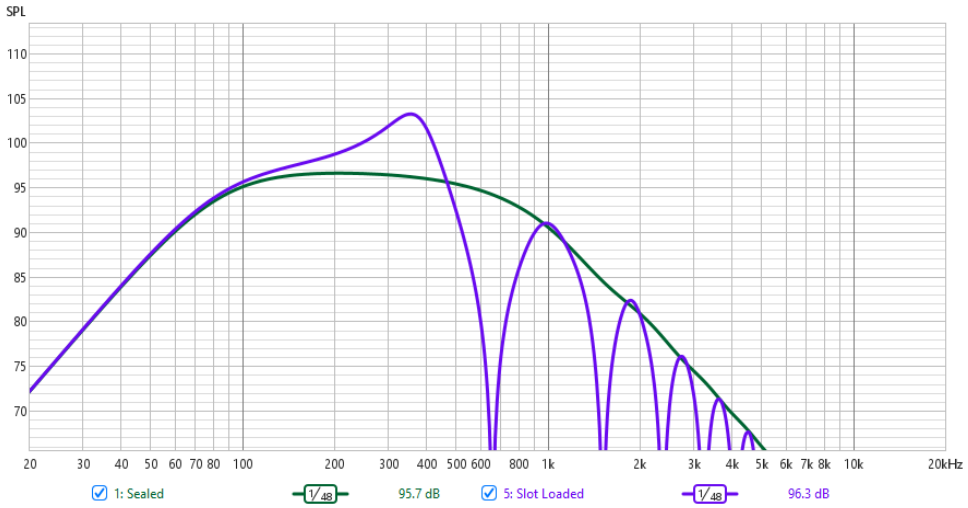


The SAWMOD features two sidefiring 10 inch low frequency transducers that share a 22 liter sealed chamber and are mounted magnet-out. This allows for superior thermal performance compared to typical sealed chamber low frequency transducers as air exchange with the motor structures can happen freely. Additionally, mounting the transducers reversed allows for a more compact arrangement as the central waveguide module occupies a significant portion of the sealed chamber. The center-center spacing of each 10” transducer is optimized to create a directional interference pattern to aid side rejection, thus lowering the horizontal directivity control cutoff. The system is acoustically crossed to the midrange section at approximately 350 Hz, a frequency chosen to balance wavelength-relative source spacing with practical enclosure constraints.

5.1 Slot-Loaded Radiation Behavior

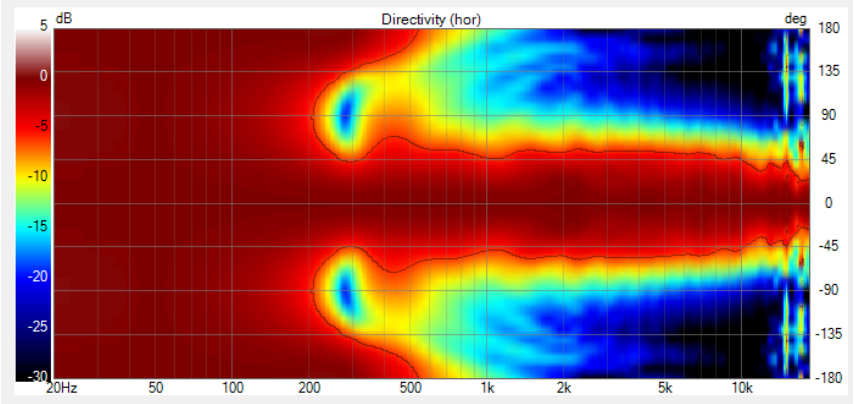
Each low-frequency driver radiates through a semi-enclosed side slot formed by the enclosure geometry. This slot-loaded configuration modifies the radiation impedance and effective loading conditions experienced by the transducers relative to a conventional sealed enclosure.

Simulation results comparing the slot-loaded configuration to a conventional sealed arrangement indicate a measurable increase in on-axis sensitivity in the upper bass and low–midrange region.



While the slot structure introduces resonant behavior characteristic of a Helmholtz system, its influence on system response is best understood as part of a coupled acoustic system rather than a simple resonator in isolation. At frequencies approaching the low–mid crossover, the spatial separation between the two low frequency drivers becomes a dominant factor in determining the horizontal radiation pattern.

Simulation of the two sources in isolation (i.e., without the slot structure) shows the expected interference pattern based on their center-to-center spacing, including the formation of lateral and rearward lobes.

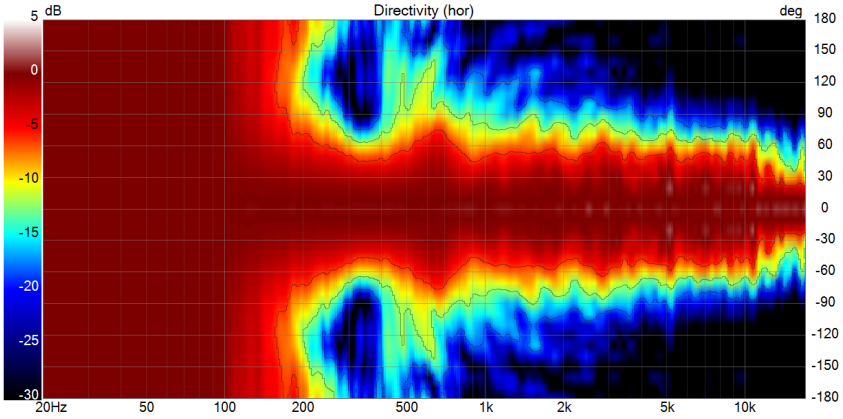


These results establish a baseline expectation for the achievable degree of side and rear rejection based purely on geometric spacing. In this configuration, the level of off-axis

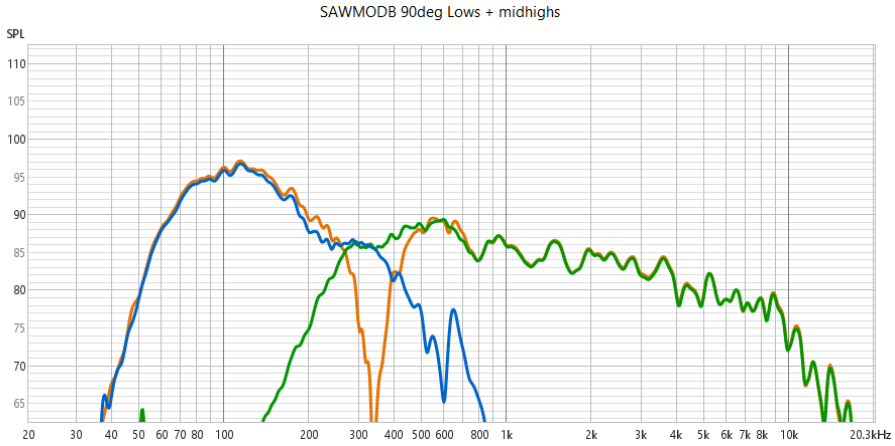
attenuation is limited by the relatively large wavelengths involved and the finite spacing between sources.

5.2 Measured Slot Behavior

Measurements of an early SAWMOD prototype reveal a significantly different radiation pattern than predicted by spacing alone. In particular, a pronounced off-axis cancellation is observed, accompanied by increased rear rejection relative to the simulated baseline.



This cancellation occurs at frequencies corresponding to the resonant behavior of the side-slot structure and is only present when both the low-frequency and midrange sections are active. When measured independently, neither the low-frequency nor midrange sections exhibit this null to the same degree. The below measurements were taken at 90 degrees off-axis illustrating the null that occurs only when both sets of drivers are playing concurrently.



The measured behavior suggests the presence of a coupled acoustic interaction between:

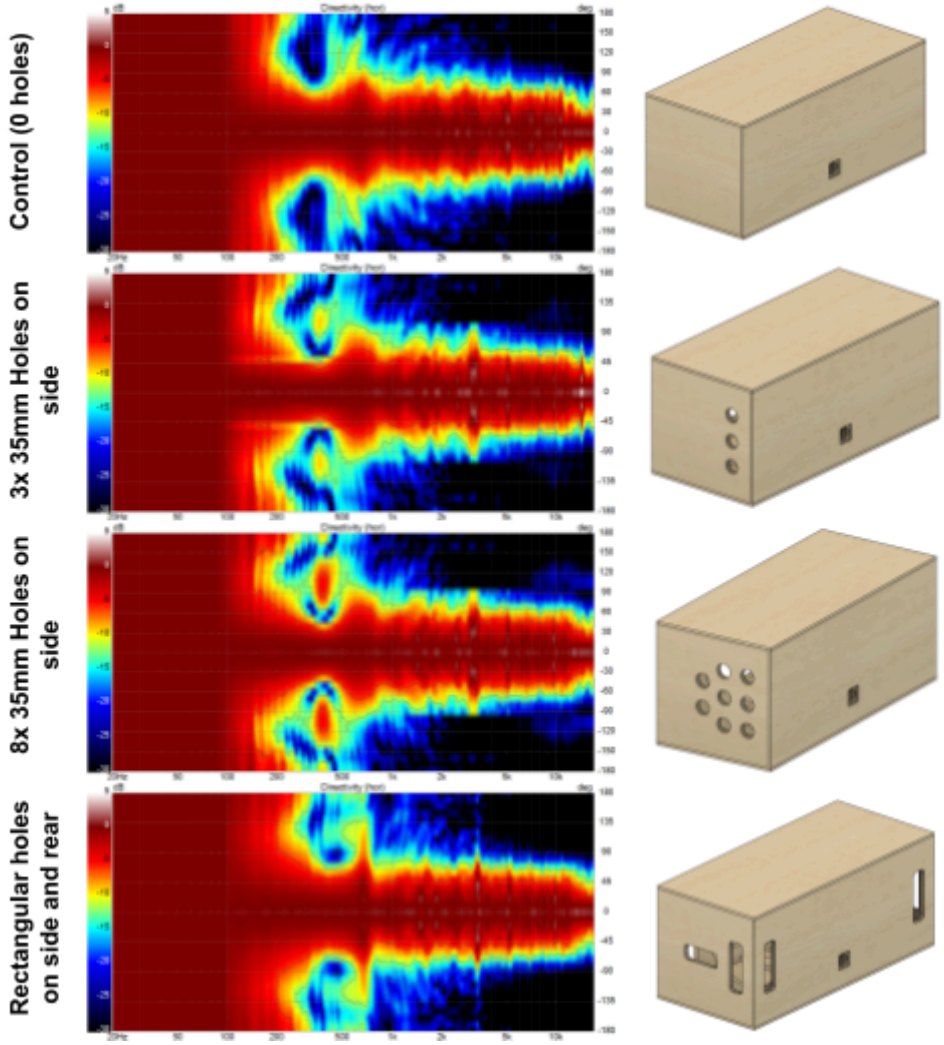
- The low-frequency transducers
- The midrange bandpass system
- The side-slot cavity structure

A plausible interpretation is that the slot volume and its associated resonance introduce a frequency-dependent phase offset between the radiated fields of the low frequency and

midrange sources. At specific angles and frequencies, this phase relationship results in enhanced destructive interference, producing the observed off-axis null.

However, this mechanism has not been fully characterized, and the interaction should be considered an observed system-level phenomenon rather than a fully modeled or optimized feature at this stage of development.

To better understand and control this behavior, a series of physical modifications were made to the side-slot structure by introducing openings (“holes”) in the slot boundaries. These modifications alter the acoustic impedance and damping characteristics of the cavity, thereby affecting its interaction with the radiated field.



This demonstrates that the slot structure plays an active role in shaping the system’s polar response and can be tuned to achieve different trade-offs between forward directivity, side rejection, and pattern smoothness.

5.3 Low Frequency Integration Summary

The low frequency section of the SAWMOD system combines spatial source arrangement and enclosure-based acoustic interaction to extend directional control below the midrange band.

Key observations include:

- The slot-loaded configuration increases acoustic efficiency in the 150-400 Hz region relative to a conventional sealed enclosure
- Source spacing alone provides a baseline level of directional control, but does not fully account for measured behavior
- Additional off-axis cancellation effects arise from coupled interactions between the low frequency drivers, midrange system, and slot cavity
- The magnitude and character of these effects can be modified through changes to the slot geometry

These results indicate that the low-frequency system operates as an integrated acoustic structure rather than a set of independent radiators, and that enclosure geometry can play a significant role in shaping system-level directivity. Further investigation is required to fully characterize and predict these interactions.

6. Conclusion

The Shared Aperture Waveguide Module (SAWMOD) is a compact multiway loudspeaker topology intended to address the limitations of conventional direct radiating, coaxial, and multiple entry horn systems. The approach centers on integrating multiple acoustic sources into a shared waveguide structure while maintaining controlled directivity and minimizing interference through critical crossover regions.

A key feature of the design is the introduction of localized contour modifications (“knuckles”) within the waveguide, enabling reduced effective source spacing for bandpass-coupled midrange drivers without compromising near-throat wavefront development. Simulation results indicate that this approach supports more consistent directivity behavior through the mid-high crossover region, while providing additional flexibility in shaping the polar response through geometric control.

The low frequency system extends this shared aperture concept by combining spatial source arrangement with enclosure-based acoustic interaction. Simulated and measured results show that the slot-loaded configuration increases efficiency in the upper bass and low midrange region, while also contributing to directional control. Measurements further indicate the presence of coupled interactions between the low frequency, midrange, and enclosure structures, producing off-axis cancellation effects not predicted by source spacing alone. These behaviors highlight the role of enclosure geometry as an active element in system-level directivity, though their underlying mechanisms require further investigation.

Future work will focus on more rigorous characterization of the observed coupling phenomena, refinement of the waveguide contouring methodology, and validation through expanded measurement and simulation studies.